

Quality-Adaptive Loss Reweighting Using Expert-Assigned Image Quality for Robust Skin Lesion Classification

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Abstract—Diagnostic image quality varies widely in clinical practice, yet quality-aware training methods have had to estimate quality from image statistics or feature norms rather than observe it, because expert judgments of image quality are almost never recorded alongside diagnostic labels. This paper has proposed quality-adaptive loss reweighting, which scales each training sample’s classification loss by a real expert-assigned rating of its diagnostic image. Two opposing directions have been evaluated: down-weighting low-quality samples as less reliable, and up-weighting them to force the model to learn from difficult images. The second, the opposite of what the closest face recognition precedents apply, has improved balanced accuracy by 0.037 and malignant sensitivity by 0.092 over an identical baseline. Because a gain of this size is comparable to the fold-to-fold standard deviation, we have validated it causally rather than statistically: a shuffled-quality control, preserving the weight distribution but permuting the lesion-to-rating assignment, has removed the gain entirely and returned performance to baseline, establishing that the improvement has depended on the quality signal’s information content rather than on generic reweighting. The full pipeline has reached 0.840 balanced accuracy, of which 65% of the total gain has been attributable to the quality-adaptive loss and the remainder to generic evaluation-time averaging. The method has been made possible by MCR-SL, the only public skin lesion dataset recording per-image expert quality ratings, on which we have also reported the first published benchmark and six robustness analyses, three of which have reversed an apparent positive result.

Index Terms—skin lesion classification, multimodal fusion, dermoscopy, image quality, metadata fusion, deep learning, medical imaging benchmark

I. INTRODUCTION

Medical images acquired in routine practice vary substantially in diagnostic quality. Lighting, focus, framing, and hair occlusion all degrade the evidence a classifier receives, and training as though every image were equally informative assumes a uniformity clinical data does not have. Face recognition has addressed this by making the objective quality aware: AdaFace [14] and MagFace [15] adapt each sample’s margin from an estimate of image quality derived from the feature norm, both reducing emphasis on difficult low-quality samples. That reliance on an estimated proxy is a necessity, not a preference, because face datasets do not record what an expert thought of each photograph.

Skin lesion classification has followed a parallel path. Convolutional networks have reached dermatologist-level accuracy on large single-modality archives [2], [3], and a smaller line of work has fused images with patient metadata [4], [5]. In neither case has a per-image expert judgment of quality been available, so quality has remained something to be inferred rather than observed.

MCR-SL [1] has removed that constraint, and is the reason the method proposed here is possible at all. It has paired 779 clinical and 1352 dermoscopic images across 240 lesions from 60 subjects with twenty-two subject-level metadata fields and, distinctively, with a one-to-ten quality rating that dermatologists have assigned to the specific image used for each lesion’s diagnosis. It is, to our knowledge, the only public skin lesion dataset carrying such a field, which makes it the enabling dataset for a loss function driven by real expert quality rather than an estimated proxy, not merely a convenient testbed for a pre-existing method.

This paper has therefore asked two separable questions: whether performance degrades on lower-quality images, and whether the training objective can use the quality signal to improve robustness. The answers have differed, the first a null result and the second a positive one validated by a causal control.

A. Research Gap

No prior work has used a direct expert rating of image quality as a loss-reweighting signal in medical image classification. Quality-aware objectives have relied on estimated proxies, quality-aware medical methods have acted on the architecture rather than the loss, and the closest loss-weighting scheme on skin lesions derives its difficulty signal from the model rather than from a human judgment of the photograph. The literature review that follows substantiates each of these in turn.

B. Contributions

This paper has made the following contributions.

- **Quality-adaptive loss reweighting:** a scheme that scales each sample’s classification loss by a real expert-assigned

image-quality rating. The direction that has worked, up-weighting low-quality samples, is the opposite of the face recognition precedents that motivated it, and we have offered a domain-specific explanation for why.

- **Causal validation via a shuffled-quality control:** permuting the lesion-to-rating assignment while preserving the weight distribution has removed the entire gain, establishing that it has depended on the quality signal’s information content and not on generic per-sample reweighting.
- **A first published benchmark on MCR-SL:** evaluated under a subject-disjoint stratified five-fold protocol, together with a sweep of eleven follow-up configurations that has isolated which interventions help at this scale and which do not.
- **Six robustness analyses, including three reversed findings:** enabled by MCR-SL’s quality ratings and partial histopathology confirmation, and reported together with the three cases in which an apparent positive finding has not survived a control.

II. LITERATURE REVIEW

Image-only dermoscopic classification has been studied extensively since Esteva *et al.* [2] matched dermatologist performance on a curated archive, with HAM10000 [3] becoming a standard benchmark. These studies have treated the image as the sole input.

Metadata fusion is a distinct thread. Pacheco and Krohling [4] have proposed the Metadata Processing Block, which reweights image features using patient metadata, evaluated on PAD-UFES-20 [5] (2298 clinical images, 1641 lesions). That dataset is the closest comparator in problem shape and is our primary external reference point, discussed later.

Quality-aware training has been developed most fully in face recognition. AdaFace [14] adapts the angular margin using the feature norm as a quality proxy, reducing emphasis on hard examples when an image is estimated to be low quality, since a degraded and difficult face is likely unidentifiable and forcing separation would fit noise. MagFace [15] ties feature magnitude to quality similarly. Both depend on an estimated proxy rather than a recorded human judgment.

Within medical imaging, quality has more often been handled architecturally. QGDR [17] routes fundus images through quality-conditioned expert branches for diabetic retinopathy grading, modifying the architecture rather than the loss, and derives its quality states largely from predicted pseudo-labels rather than from expert annotation.

Closest to this work, Hu and Yang [16] have applied knowledge-based loss weighting on HAM10000, assigning weights at the class level, favouring minority classes, and at the instance level, favouring challenging samples. Their instance-level term shares this paper’s direction in emphasising difficult examples, but its notion of difficulty is derived from the model and data themselves. The signal used here is external to both: an expert’s judgment of the photograph, recorded independently of how any model performs on it. Our fusion

module draws on Squeeze-and-Excitation [6] and is used as an established component, not claimed as a contribution.

III. PROPOSED METHODOLOGY

Fig. 1 summarizes the architecture. This paper has proposed a channel-gated fusion method for cross-modal skin lesion classification and has compared it against two baselines under an identical training and evaluation protocol.

A. Image Encoder

The upper stream of Fig. 1 encodes the dermoscopic image. An ImageNet-pretrained EfficientNet-B0 [7], [8] is trained end to end with no frozen layers, and a single forward pass returns both the final convolutional feature map, of size $1280 \times 7 \times 7$, and its globally average-pooled 1280-dimensional summary. The feature map is what the fusion stage gates; the pooled vector is used only by the concatenation baseline.

B. Metadata Encoder

The lower stream encodes patient and lesion context. Seventeen categorical fields each receive their own embedding of twelve dimensions, sized to the field’s cardinality plus one reserved index for missing values, so missingness is represented explicitly and never imputed. Four numeric fields are z-scored using training-fold statistics only, and each is paired with a binary indicator recording whether it was present. Concatenating these gives a 212-dimensional vector, which a two-layer perceptron maps to the 128-dimensional metadata representation.

C. Channel-Gated Fusion

Two fusion variants share a common 256-dimensional projection. The late-fusion baseline simply concatenates the pooled image vector with the metadata vector. The proposed channel-gated variant, drawn as $\otimes$ in Fig. 1, instead maps the metadata vector through a linear layer and a sigmoid to a 1280-dimensional gate in $[0, 1]$, multiplies the convolutional feature map channel by channel by that gate, and only then pools. Patient context therefore modulates which visual channels survive pooling rather than being appended after it. This is the Squeeze-and-Excitation mechanism [6] conditioned on metadata instead of on the feature map’s own statistics, and it is used here as an established component rather than claimed as a contribution.

D. Prediction Heads

Two heads operate on the fused 256-dimensional representation. The binary head is a single linear layer followed by a sigmoid, giving $P(\text{malignant})$, and is the only head whose output is reported as a headline result. A nine-class auxiliary head predicts the unified diagnosis under a class-weighted cross-entropy contributing at $0.4\times$, and is reported as an exploratory table only, since several of its classes contain fewer than ten lesions.

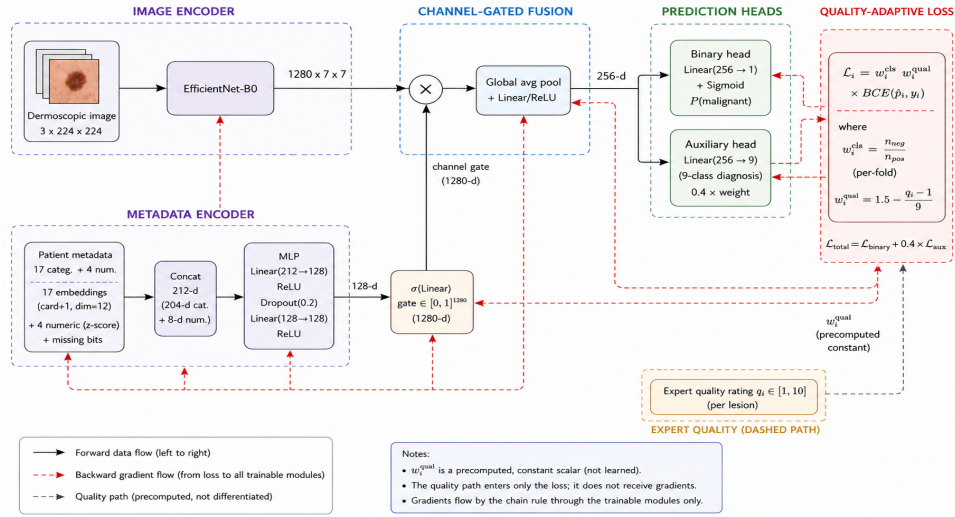


Fig. 1. Overview of the proposed multimodal architecture and training objective. The image and metadata streams are encoded separately and meet at a metadata-conditioned channel gate, where a sigmoid gate rescales the convolutional feature map channel by channel before global average pooling and projection. The binary and auxiliary prediction heads operate on the resulting 256-dimensional representation. Solid arrows denote the forward data flow, while the red backward path illustrates gradient propagation from the loss through the trainable network components. The expert quality rating enters only through the dashed quality path; w_i^{qual} is a precomputed constant and is not learned or differentiated.

E. Quality-Adaptive Loss

The paper’s methodological contribution modifies neither the architecture nor the class of loss function, but the weight each sample carries within the existing objective. Let $q_i \in [1, 10]$ be the mean expert quality rating of lesion i ’s diagnostic image. A per-sample multiplicative weight w_i^{qual} has been applied alongside the existing per-fold class weight w_i^{cls} :

$$\mathcal{L}_i = w_i^{\text{cls}} \cdot w_i^{\text{qual}} \cdot \text{BCE}(\hat{y}_i, y_i). \quad (1)$$

Two opposing directions have been evaluated, each mapping the rating range onto $[0.5, 1.5]$:

$$\text{trust: } w_i^{\text{qual}} = 0.5 + (q_i - 1)/9, \quad (2)$$

$$\text{hard_mining: } w_i^{\text{qual}} = 1.5 - (q_i - 1)/9. \quad (3)$$

The *trust* direction has treated low-quality images as less reliable and down-weighted them. The *hard_mining* direction has done the opposite, forcing the model to work harder on degraded images. Lesions without a valid rating have received a neutral weight of 1.0. The range has been fixed rather than searched, and only these two directions have been evaluated, so that the comparison isolates direction rather than magnitude. Nothing unusual happens to the gradient: w_i^{qual} is a precomputed, non-differentiable scalar coefficient on the loss, not a learned parameter, so gradients propagate through the shared architecture by the standard chain rule from the weighted binary term. The mechanism changes only how much each sample contributes, never what is differentiated.

Only *hard_mining* has survived validation, as shown below, and its direction is the opposite of AdaFace’s and MagFace’s [14], [15]. One possible reason is that the two settings degrade differently. In face identification a sufficiently degraded photograph can make the identity unrecoverable, so emphasizing

such a sample risks fitting noise. In dermoscopy the diagnostic label was in fact assigned from that same image, so the label remains valid and the sample remains an informative hard example.

This mechanism is distinct from the auxiliary quality-regression head noted earlier, which has instead asked the model to *predict* the rating as a secondary output. That variant adds a competing objective to a shared representation, whereas reweighting adds no parameters and changes only which samples dominate the gradient. The two are mechanistically different interventions, and the auxiliary head’s negative result is reported separately and remains valid; it has not been superseded by the reweighting result.

F. Training and Evaluation Protocol

Every configuration has been evaluated under subject-disjoint stratified five-fold cross-validation, assigning each subject’s lesions to exactly one fold so that no subject or near-duplicate lesion image has appeared in both a training and an evaluation fold. Folds have been balanced by greedily assigning subjects to minimize the difference in malignant lesion count across folds. For each of the five rotations, one fold has been held out as the reported test fold and a second, disjoint fold has been held out purely for checkpoint selection by validation balanced accuracy; the test fold has never influenced model selection, and no hyperparameter has been tuned against final test-fold results.

Seven follow-up interventions have additionally been evaluated on the channel-gated method under the identical protocol: focal loss [12], dermoscopy preprocessing (hair removal and gray-world color normalization), a supervised contrastive auxiliary loss [11], an alternative optimizer using decoupled weight decay with a cosine schedule [10], sharpness-aware

minimization [9], test-time augmentation by horizontal flip averaging, and averaging predictions across every dermoscopic image of a lesion. The last two are evaluation-time only. Four stacked combinations bring the total to eleven configurations.

IV. EXPERIMENTAL RESULTS

A. Dataset

MCR-SL [1] comprises 240 lesions from 60 subjects, 42 malignant and 192 non-malignant, with 1352 dermoscopic and 779 clinical images, 28 lesions histopathology-confirmed, and 16 categorical plus 4 numeric metadata fields. Six lesions carry an undetermined malignancy label and are excluded, leaving 234 usable; The evaluation study below traces the further reduction to the 231 used in the analyses.

B. Evaluation Metrics

Six metrics are reported for every configuration: accuracy, sensitivity, specificity, macro-averaged F1, balanced accuracy, and the area under the receiver operating characteristic curve (AUROC). Each is aggregated across the five test folds as a mean and standard deviation.

Accuracy is the proportion of all lesions classified correctly. On its own it is a weak summary here, because predicting every lesion non-malignant would already score above 80% at this class balance.

Sensitivity is the proportion of malignant lesions correctly identified. It is the clinically decisive measure, since a missed malignancy is a missed cancer, and it is the metric the proposed loss is intended to improve.

Specificity is the proportion of non-malignant lesions correctly cleared. Low specificity means unnecessary biopsies and avoidable patient anxiety, so it constrains how far sensitivity can be bought.

Macro-averaged F1 is the unweighted mean of the per-class harmonic means of precision and recall,

$$\text{macro-F1} = \frac{1}{2} \sum_{c \in \{0,1\}} \frac{2 \text{Pre}_c \text{Rec}_c}{\text{Pre}_c + \text{Rec}_c}, \quad (4)$$

and rewards doing well on both classes rather than on the majority one alone.

Balanced accuracy is the unweighted mean of the two class-wise rates,

$$\text{BAcc} = \frac{1}{2} (\text{Sen} + \text{Spe}), \quad (5)$$

and is used both for checkpoint selection and as the primary comparison metric, since it corrects for the malignant class being a 17.9% minority.

AUROC is the probability that a randomly drawn malignant lesion receives a higher predicted probability than a randomly drawn non-malignant one. It is threshold-independent, so it measures ranking quality separately from where the decision boundary happens to sit.

C. Existing vs. Proposed Methods

This subsection compares the proposed method against the existing approaches internally: every configuration in Table I has been trained and evaluated by us under the identical protocol, on the same folds and the same lesions, so the differences are attributable to the method alone. Values are given to three decimals, since several are too close for two decimals to separate. Channel-gated fusion has reached the highest macro-F1 but has not dominated: the image-only baseline has higher sensitivity (0.697 vs. 0.672) and AUROC (0.895 vs. 0.881), and its balanced accuracy is effectively tied (0.784 vs. 0.783, against a standard deviation of 0.07–0.09). We report this rather than framing it as a clean win for the fusion method.

The quality-aware variant, which adds an auxiliary head predicting the quality rating, has underperformed on five of six metrics: balanced accuracy 0.740 vs. 0.783 and sensitivity 0.578 vs. 0.672, against a small specificity gain (0.902 vs. 0.895). It has become more conservative, trading a large sensitivity loss for a small specificity gain, the worse direction for cancer screening. This is a negative result for *predicting* quality, and is mechanistically distinct from the reweighting scheme introduced earlier.

TABLE I
RESULTS OF EXISTING AND PROPOSED METHODS (MEAN $\pm$ STD, FIVE FOLDS). BOLD MARKS THE BEST VALUE IN EACH ROW

Metric	Image-only	Late fusion	Channel-gated
Accuracy	0.813 $\pm$ 0.020	0.831 $\pm$ 0.052	0.833 $\pm$ 0.070
Balanced acc.	0.784 $\pm$ 0.073	0.768 $\pm$ 0.090	0.783 $\pm$ 0.085
Macro F1	0.733 $\pm$ 0.038	0.743 $\pm$ 0.051	0.757 $\pm$ 0.073
Sensitivity	0.697 $\pm$ 0.187	0.644 $\pm$ 0.191	0.672 $\pm$ 0.159
Specificity	0.872 $\pm$ 0.061	0.891 $\pm$ 0.021	0.895 $\pm$ 0.048
AUROC	0.895 $\pm$ 0.053	0.851 $\pm$ 0.080	0.881 $\pm$ 0.065

Channel-gated, quality-aware variant: accuracy 0.812 $\pm$ 0.068, balanced accuracy 0.740 $\pm$ 0.065, macro F1 0.715 $\pm$ 0.065, sensitivity 0.578 $\pm$ 0.177, specificity 0.902 $\pm$ 0.073, AUROC 0.879 $\pm$ 0.041.

Both weighting directions introduced earlier have been trained under the identical protocol, so the loss weighting is the only difference between them and the baseline. The *hard_mining* direction has improved every core metric: balanced accuracy by 0.037 (0.783 to 0.820), sensitivity by 0.092 (0.672 to 0.764), AUROC by 0.025 (0.881 to 0.906) and accuracy to 0.845, against a small specificity cost (0.895 to 0.875). The sensitivity gain is the clinically weightier one. The *trust* direction is approximately flat on balanced accuracy (0.788) and is not treated as an improvement. Since 0.037 is smaller than the fold-to-fold standard deviation of roughly 0.08, the comparison is underpowered, and we report that plainly: across the five paired folds *hard_mining* wins three on balanced accuracy (Wilcoxon $p = 0.81$) and four on AUROC ($p = 0.31$), and on sensitivity wins two while tying three, never losing a fold ($p = 0.18$). None approaches significance at $N = 5$, so the claim does not rest on this comparison; the control below tests the mechanism causally instead.

Causal validation via a shuffled-quality control. The obvious objection is that any per-sample reweighting might act

as a regularizer, leaving the ratings incidental. We have tested this directly: within each fold’s training portion the ratings are permuted across lesions, preserving the weight distribution exactly while destroying the lesion-to-rating correspondence. All else is identical, and evaluation always uses the true ratings.

Under this control, *hard_mining*’s balanced accuracy falls from 0.820 to 0.775 and its sensitivity from 0.764 to 0.667, both returning to the baseline’s level (0.783 and 0.672).

A distribution-matched but uninformative weighting therefore recovers none of the gain, which establishes that the improvement depends on the weights tracking real quality rather than on reweighting as such. This is a causal argument rather than a statistical one, and it is why the result is reported despite an effect size comparable to the fold-to-fold standard deviation.

The control has also been informative negatively. The *trust* variant’s one apparent advantage, a narrower accuracy gap between the highest and lowest quality terciles (0.082 against the baseline’s 0.091), narrows further still under shuffling, to 0.021, while its balanced accuracy falls from 0.788 to 0.763: a weighting carrying no quality information produces a flatter profile than the mechanism intended to produce one. That advantage is therefore reported as a reversed finding rather than a second positive result.

D. Comparison with the Literature

Unlike the internal comparison above, this subsection places the results against work by other authors, trained by them on a different dataset. No configuration here was retrained by us, and there is no shared test set, so Table II is context rather than a controlled comparison.

TABLE II
EXTERNAL COMPARISON AGAINST THE CLOSEST PUBLISHED
BINARY-TASK RESULT. NOT A CONTROLLED COMPARISON: DIFFERENT
DATASET, DIFFERENT AUTHORS, NO SHARED TEST SET

Source	Task	N	Bal. acc.
This paper, channel-gated	binary	234 lesions	0.783
This paper, + <i>hard_mining</i>	binary	234 lesions	0.820
This paper, best configuration	binary	234 lesions	0.840
DiffMIC, PAD-UFES-20 [13]	binary	2298 images	0.836

PAD-UFES-20’s six-class balanced accuracies (0.765–0.842) are not valid comparators for a binary task despite the shared metric name. The binary result we have verified directly is Uliana and Krohling’s [13], 0.836 from a diffusion classifier over 2298 images under five-fold cross-validation, at the image level rather than our lesion level. This paper’s best configuration has reached 0.840 ± 0.095 . The nominal 0.004 difference is more than twenty times smaller than our own fold-to-fold standard deviation, so the two results are statistically indistinguishable and we have described ours as within noise of that benchmark rather than as exceeding it. The comparison is also not like for like: the two counts differ in unit, our 234 lesions against their 2298 images, and the underlying datasets differ by roughly sevenfold in lesion count and, more consequentially, in class balance, 17.9% malignant

here against 47.5% there. HAM10000 [3] headline results have reached higher raw accuracy, but on a dermoscopy-only task with no metadata and roughly forty times more training images, and have not been treated as comparable.

MCR-SL has had no prior published benchmark, so this result is the current best on it by default; we have avoided framing that as a state-of-the-art claim, since there has been no prior number to surpass, and have emphasized the loss mechanism and its shuffled-quality validation as the principal contribution.

E. Ablation Study

Eleven configurations have been evaluated under the identical protocol, each varying one component of the channel-gated method: the loss (focal loss [12], the *trust* direction), the input (hair removal and colour normalisation), the objective (a contrastive auxiliary term [11], the quality-regression head), the optimiser (decoupled weight decay with a cosine schedule [10], sharpness-aware minimisation [9]), and the evaluation (flip-averaged and multi-image test-time averaging), plus four stacked combinations. Sharpness-aware minimisation with test-time augmentation reached 0.822, the best of this sweep. Contrastive pretraining, the alternative optimiser and the quality-regression head each underperformed the baseline, and stacking every promising intervention together reached only 0.759.

Applying the two evaluation-time interventions on top of *hard_mining*, rather than on top of the plain baseline, has produced this paper’s best configuration. Table III decomposes it. Test-time augmentation has raised balanced accuracy to 0.833 and sensitivity to 0.808, the highest sensitivity of any configuration reported here; adding multi-image averaging has raised balanced accuracy further to 0.840 ± 0.095 . Notably, multi-image averaging has stacked cleanly here, whereas it reduced balanced accuracy when combined with sharpness-aware minimization and test-time augmentation in the sweep above; that non-additivity has therefore been specific to that combination rather than a general property of the intervention.

The decomposition matters for how the headline number should be read. Of the total 0.057 gain over the baseline, 0.037, or roughly 65%, has come from the quality-adaptive loss and has been causally validated by the shuffled-quality control above. The remaining 0.020 has come from generic evaluation-time averaging that requires no retraining and has nothing to do with image quality. We state this split explicitly so that the contribution attributable to the proposed mechanism is not overstated by the final figure.

F. Evaluation Study

These analyses use the channel-gated out-of-fold predictions over the 231 lesions carrying both a prediction and a quality rating; three of the 234 lacked a usable image in their assigned test fold and were excluded rather than imputed. Table IV collects the numbers, and the items below give only the interpretation.

TABLE III
DECOMPOSITION OF THE BEST CONFIGURATION. ONLY THE FIRST STEP
USES THE QUALITY SIGNAL; THE REMAINDER IS GENERIC
EVALUATION-TIME AVERAGING. BOLD MARKS THE BEST VALUE IN EACH
COLUMN

Step	Quality?	Bal. acc.	Δ	Sens.
Channel-gated baseline	no	0.783	—	0.672
+ <i>hard_mining</i> loss	yes	0.820	+0.037	0.764
+ test-time augmentation	no	0.833	+0.050	0.808
+ multi-image averaging	no	0.840	+0.057	0.783
<i>hard_mining</i> , shuffled	control	0.775	−0.008	0.667

TABLE IV
EVALUATION STUDY OVER THE 231 EVALUATED LESIONS. TERCILE SIZES
ARE UNEVEN BECAUSE THE MEAN OF UP TO THREE INTEGER RATINGS
TAKES FEW DISTINCT VALUES

Accuracy by expert quality tercile			
Tercile	<i>N</i>	Baseline	<i>hard_mining</i>
Low	85	0.776	0.776
Mid	93	0.914	0.903
High	53	0.868	0.868
Malignant sensitivity, low tercile	23	0.667	0.810
Certainty vs. error (Spearman ρ)			
Stratum	<i>N</i>	ρ	<i>p</i>
Pooled	231	−0.175	0.008
Non-malignant only	189	−0.127	0.083
Malignant only	42	−0.093	0.559
Ground-truth source and intra-subject consistency			
Group	<i>N</i>	Accuracy	
Histopathology-confirmed	28	0.750	
Panel-consensus only	203	0.867	
Consistent subjects (null 29.9)	55	27, $p = 0.954$	

Quality-stratified performance. The pattern across terciles is not monotonic, and the Spearman correlation between rating and error is -0.093 ($p = 0.158$), so no strong quality-performance relationship is detected at this sample size, neither confirming nor ruling one out. The validated *hard_mining* configuration does not flatten the profile either; its benefit appears within terciles, most visibly as a large sensitivity gain on the lowest-quality one.

Reversed findings. Three apparently positive results have not survived scrutiny and are reported rather than omitted. *Trust*’s narrower quality-tercile gap was undercut by the shuffled control narrowing it further, as discussed above. The pooled certainty-error correlation dissolves once stratified by class, having reflected class composition rather than an independent relationship. And a proposed change to train on all images per lesion proved on verification to describe what the pipeline already did. At this sample size an uncontrolled first look repeatedly produces positives that controls dissolve, and reporting that pattern is part of the contribution.

Intra-subject consistency. Errors do not concentrate in particular patients: fewer subjects are perfectly consistent than the permutation null predicts, and no subject has all of its lesions misclassified.

Ground-truth source. The model performs worse on histopathology-confirmed lesions than on panel-consensus ones. A likely explanation is that histopathology is ordered more often for clinically ambiguous lesions, making that subset harder by construction. With 28 lesions this is treated as qualitative only.

Qualitative analysis. Gradients have been taken with respect to the feature map *after* the metadata gate, the representation the decision is computed from, with the metadata branch held fixed. Fig. 2 compares the baseline and reweighted models on the two lowest-quality malignant lesions the loss has flipped from a miss to a correct call. In the first, a basal cell carcinoma rated 5.7, baseline attention is spread diffusely and the lesion is confidently missed; under reweighting attention contracts onto a single focus and the call becomes correct. In the second, a melanoma rated 4.3, baseline attention falls on the image edge away from the pigmented region; under reweighting it moves onto the lesion. Both lie in the lowest quality tercile, where reweighting assigns its largest weights. The maps are 7×7 upsampled and indicate approximate attention only. These are illustrative rather than representative: across the 231 lesions the reweighting fixes six malignant lesions and breaks two, a net gain of four, which is what a $+0.092$ sensitivity amounts to at this scale.

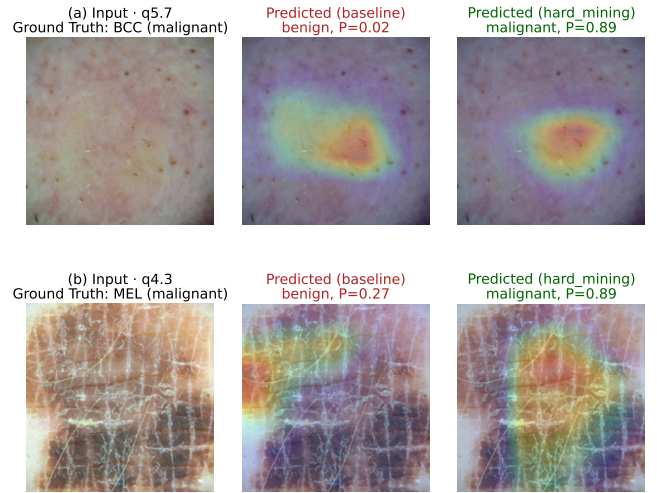


Fig. 2. Grad-CAM on the post-gate feature map for two lesions that quality-adaptive reweighting flips from a miss to a correct call.

V. CONCLUSION AND FUTURE WORK

This paper has proposed quality-adaptive loss reweighting, scaling each training sample’s classification loss by a real expert-assigned rating of its diagnostic image. Up-weighting low-quality samples has improved balanced accuracy by 0.037, sensitivity by 0.092 and AUROC by 0.025 over an identical baseline, and evaluation-time averaging on top has reached 0.840 ± 0.095 ; of that 0.057 total gain, 65% is attributable to the loss and the remainder to generic technique. The validated direction is the opposite of the face recognition precedents that motivated it [14], [15], likely because a dermatologist can usually still diagnose from a degraded photograph while a degraded face can lose its identity entirely.

The effect size deserves care. A gain of 0.037 is comparable to the fold-to-fold standard deviation of roughly 0.08, and no paired fold-level test reaches significance at $N = 5$. The claim

rests instead on the shuffled-quality control: reweighting by permuted ratings returns performance to baseline. We regard causal evidence as the appropriate standard at this scale, and have reported the effect size plainly rather than letting the final figure carry the claim. Two questions have been answered differently: performance shows no significant relationship with expert-rated quality, a null that also held for expert certainty once class composition was controlled, yet the objective has still exploited that signal.

The study is limited by MCR-SL's size, 234 usable lesions from one institution with 28 histopathology-confirmed, by a single training seed, and by a 17.9% malignant rate leaving eight or nine malignant lesions per test fold. Future work should repeat the control over many permutations for an empirical p -value, and validate the mechanism on other quality-labelled medical datasets, most directly DeepDRiD [18].

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